ALGEBRA 2 · UNIT 5: RADICALS & RATIONAL EXPONENTS

Crash Scene Investigator

Use radical formulas and rational exponents to reconstruct an accident and study motion.

CP1

2–3 class periods

Name: _____ Date: _____ Period: _____

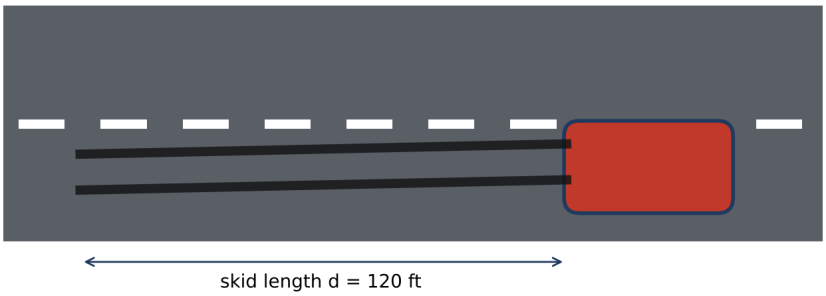
The Scenario

A car skidded 120 feet on a dry road before stopping in a 45-mph zone. Accident investigators use s = √(30df), where s is speed in mph, d is skid length in feet, and f is the road's friction factor (dry asphalt 0.75, wet asphalt 0.45).

Your lab will estimate speeds, solve for unknowns, and explore other formulas with radicals and rational exponents, then write a report.

Crash Lab Formula

speed $s = \sqrt{(30 \cdot d \cdot f)}$ (mph; d in feet; f = road friction factor)



Learning Goals

- I can rewrite between radicals and rational exponents.
- I can solve and rearrange radical formulas.
- I can graph square-root functions as transformations.

Project Timeline

When	What you will do
Day 1	The Skid Marks
Day 2	Solve & Graph
Day 3	Orbits & Report
Final	Investigation Report + reflection

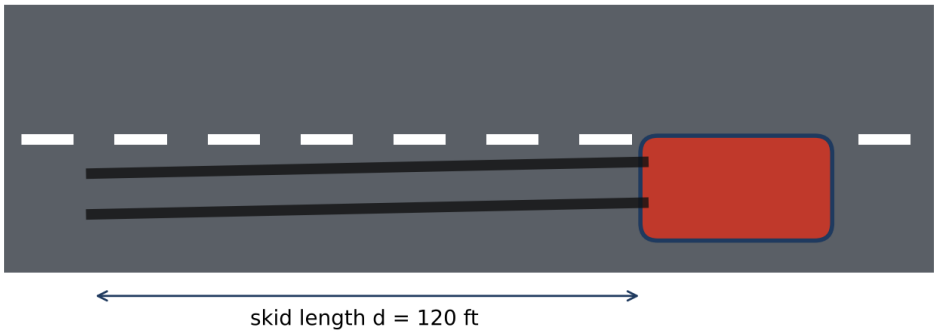
Part 1: The Skid Marks

Day 1 · Use s = √(30df).

1. Find the dry-road speed for a 120-ft skid in simplest radical form. Then write the formula using a rational exponent.

Crash Lab Formula

speed s = √(30 · d · f) (mph; d in feet; f = road friction factor)



2. Solve the formula for d. Use it to find the skid length at 45 mph and at 60 mph on dry asphalt.

3. A second skid is 4 times as long on the same road. Show algebraically that the speed is 2 times as large.

Part 2: Solve & Graph

Day 2 · Check every solution.

4. Solve $\sqrt{(x + 7)} = x - 5$ and identify any extraneous solution.

5. Solve $\sqrt[3]{(2x - 3)} = 3$.

6. Describe $s(d) = \sqrt{(22.5d)}$ as a transformation of $y = \sqrt{d}$. Graph both.

7. On a wet road, $s(d) = \sqrt{(13.5d)}$. Is this graph above or below the dry graph? Why?

Part 3: Orbits & Report

Day 3 · Kepler: $T^2 = a^3$, where T is the orbit time in years and a is the average distance from the Sun in AU. So $T = a^{1.5}$.

8. Mars has a = 1.52 AU. Find its orbit time.

9. Saturn's orbit takes 29.5 years. Solve for a (use the ⅔ power) and find a.

10. Write the case report and include one radical and one rational-exponent result.

Final Product: Investigation Report

Your final product must include:

- Skid analysis with a rearranged formula
- Solved radical equations
- Graphs and transformations
- Orbit calculations
- Report

Reflection

R1. Why is $a^{1.5}$ the same as $a\sqrt{a}$?

R2. How does the friction factor change the graph?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Conclusions are supported by correct radical calculations, and any extraneous solutions are identified. Goes beyond the minimum with a creative, realistic choice.	Conclusions are supported by correct radical calculations, and any extraneous solutions are identified.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Report includes speed estimates with simplified radicals, solved formulas, a graph, and a conclusion about the case. Polished, organized, and visually engaging.	Report includes speed estimates with simplified radicals, solved formulas, a graph, and a conclusion about the case. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.